

Equation	Closed-Form Expression	Variable domains	Variables	Description	Taxonomy (Domain / Subdomain / Specific)	SRSD Hard
I.6.2a	$\frac{\sqrt{2}e^{-\frac{\theta^2}{2}}}{2\sqrt{\pi}}$	$\theta \in [1.0, 3.0]$	θ : angle/position	Gaussian distribution (normal distribution with $\sigma = 1$)	Mathematics / Probability Theory / Gaussian Distribution	✓
I.6.2	$\frac{\sqrt{2}e^{-\frac{\theta^2}{2\sigma^2}}}{2\sqrt{\pi}\sigma}$	$\sigma \in [1.0, 3.0]$ $\theta \in [1.0, 3.0]$	σ : std deviation θ : variable	Gaussian/normal distribution probability density	Mathematics / Probability Theory / Gaussian Distribution	✓
I.6.2b	$\frac{\sqrt{2}e^{-\frac{(\theta-\theta_1)^2}{2\sigma^2}}}{2\sqrt{\pi}\sigma}$	$\sigma \in [1.0, 3.0]$ $\theta \in [1.0, 3.0]$ $\theta_1 \in [1.0, 3.0]$	σ : std deviation θ : variable θ_1 : mean	Gaussian distribution with mean θ_1	Mathematics / Probability Theory / Gaussian Distribution	✓
I.8.14	$\sqrt{(-x_1+x_2)^2+(-y_1+y_2)^2}$	$x1 \in [1.0, 5.0]$ $x2 \in [1.0, 5.0]$ $y1 \in [1.0, 5.0]$ $y2 \in [1.0, 5.0]$	x_1 : x-coordinate of point 1 x_2 : x-coordinate of point 2 y_1 : y-coordinate of point 1 y_2 : y-coordinate of point 2	Euclidean distance between two points in 2D	Mathematics / Geometry / Euclidean Distance	
I.9.18	$\frac{Gm_1m_2}{(-x_1+x_2)^2+(-y_1+y_2)^2+(-z_1+z_2)^2}$	$m1 \in [1.0, 2.0]$ $m2 \in [1.0, 2.0]$ $G \in [1.0, 2.0]$ $x1 \in [3.0, 4.0]$ $x2 \in [1.0, 2.0]$ $y1 \in [3.0, 4.0]$ $y2 \in [1.0, 2.0]$ $z1 \in [3.0, 4.0]$ $z2 \in [1.0, 2.0]$	m_1 : mass of body 1 m_2 : mass of body 2 G : gravitational constant x_1 : x-coordinate of body 1 x_2 : x-coordinate of body 2 y_1 : y-coordinate of body 1 y_2 : y-coordinate of body 2 z_1 : z-coordinate of body 1 z_2 : z-coordinate of body 2	Newton's law of gravitation (gravitational potential energy)	Physics / Classical Mechanics / Gravitational Force	✓
I.10.7	$\frac{m_0}{\sqrt{1-\frac{v^2}{c^2}}}$	$m_0 \in [1.0, 5.0]$ $v \in [1.0, 2.0]$ $c \in [3.0, 10.0]$	m_0 : rest mass v : velocity c : speed of light	Relativistic mass (special relativity)	Physics / Relativity / Special Relativity	
I.11.19	$x_1y_1+x_2y_2+x_3y_3$	$x1 \in [1.0, 5.0]$ $x2 \in [1.0, 5.0]$ $x3 \in [1.0, 5.0]$ $y1 \in [1.0, 5.0]$ $y2 \in [1.0, 5.0]$ $y3 \in [1.0, 5.0]$	x_1 : 1st component of vector x x_2 : 2nd component of vector x x_3 : 3rd component of vector x y_1 : 1st component of vector y y_2 : 2nd component of vector y y_3 : 3rd component of vector y	Dot product of two 3D vectors	Mathematics / Linear Algebra / Vector Operations	
I.12.1	$Nn\mu$	$mu \in [1.0, 5.0]$ $Nn \in [1.0, 5.0]$	Nn : number density μ : magnetic moment	Total magnetic moment (dipole moment)	Physics / Electromagnetism / Magnetic Dipole	
I.12.2	$\frac{q_1q_2}{4\pi\epsilon r^2}$	$q1 \in [1.0, 5.0]$ $q2 \in [1.0, 5.0]$ $\epsilon \in [1.0, 5.0]$ $r \in [1.0, 5.0]$	q_1 : charge of particle 1 q_2 : charge of particle 2 ϵ : permittivity of medium r : distance between charges	Coulomb's law (electrostatic force)	Physics / Electromagnetism / Electrostatics	
I.12.4	$\frac{q1}{4\pi\epsilon r^2}$	$q1 \in [1.0, 5.0]$ $\epsilon \in [1.0, 5.0]$ $r \in [1.0, 5.0]$	q_1 : charge ϵ : permittivity r : distance	Electric field magnitude from point charge	Physics / Electromagnetism / Electric Field	

I.12.5	Efq_2	$q_2 \in [1.0, 5.0]$	Ef : electric field	Electric force on charge in field	Physics / Electromagnetism / Electric Force
I.12.11	$q(Bv \sin(\theta) + Ef)$	$Ef \in [1.0, 5.0]$ $q \in [1.0, 5.0]$	q_2 : charge q : charge	Lorentz force (combined electric and magnetic)	Physics / Electromagnetism / Lorentz Force
I.13.4	$\frac{m(u^2 + v^2 + w^2)}{2}$	$Ef \in [1.0, 5.0]$ $B \in [1.0, 5.0]$ $v \in [1.0, 5.0]$ $\theta \in [1.0, 5.0]$	Ef : electric field B : magnetic field v : velocity θ : angle	Kinetic energy in 3D	Physics / Classical Mechanics / Kinetic Energy
I.13.12	$Gm_1m_2 \cdot \left(\frac{1}{r_2} - \frac{1}{r_1}\right)$	$m \in [1.0, 5.0]$ $u \in [1.0, 5.0]$ $v \in [1.0, 5.0]$ $w \in [1.0, 5.0]$	m : mass of particle u : velocity x-component v : velocity y-component w : velocity z-component	Gravitational potential energy difference	Physics / Classical Mechanics / Gravitational Potential
I.14.3	gmz	$G \in [1.0, 5.0]$ $m1 \in [1.0, 5.0]$ $m2 \in [1.0, 5.0]$ $r1 \in [1.0, 5.0]$ $r2 \in [1.0, 5.0]$	G : gravitational constant m_1 : mass of body 1 m_2 : mass of body 2 r_1 : initial radial distance r_2 : final radial distance	Gravitational potential energy	Physics / Classical Mechanics / Gravitational Potential
I.14.4	$\frac{k_{spring}x^2}{2}$	$m \in [1.0, 5.0]$ $g \in [1.0, 5.0]$ $z \in [1.0, 5.0]$	m : mass g : grav acceleration z : height	Gravitational potential energy	Physics / Classical Mechanics / Gravitational Potential
I.14.4	$\frac{k_{spring}x^2}{2}$	$k_{spring} \in [1.0, 5.0]$	k_{spring} : spring constant	Elastic potential energy (Hooke's law)	Physics / Classical Mechanics / Elastic Energy
I.15.3x	$\frac{-tu+x}{\sqrt{1-\frac{u^2}{c^2}}}$	$x \in [1.0, 5.0]$ $x \in [5.0, 10.0]$	x : displacement x : position	Lorentz transformation for position (special relativity)	Physics / Relativity / Lorentz Transformation ✓
I.15.3t	$\frac{t-\frac{ux}{c^2}}{\sqrt{1-\frac{u^2}{c^2}}}$	$u \in [1.0, 2.0]$ $c \in [3.0, 20.0]$ $t \in [1.0, 2.0]$	t : time u : relative velocity c : speed of light	Lorentz transformation for time (special relativity)	Physics / Relativity / Lorentz Transformation ✓
I.15.1	$\frac{m_0v}{\sqrt{1-\frac{v^2}{c^2}}}$	$x \in [1.0, 5.0]$ $c \in [3.0, 10.0]$ $u \in [1.0, 2.0]$ $t \in [1.0, 5.0]$	t : time x : position u : relative velocity c : speed of light	Lorentz transformation for time (special relativity)	Physics / Relativity / Lorentz Transformation ✓
I.15.1	$\frac{m_0v}{\sqrt{1-\frac{v^2}{c^2}}}$	$m_0 \in [1.0, 5.0]$	m_0 : rest mass	Relativistic momentum	Physics / Relativity / Special Relativity
I.16.6	$\frac{u+v}{1+\frac{uv}{c^2}}$	$v \in [1.0, 2.0]$ $c \in [3.0, 10.0]$	v : velocity c : speed of light	Relativistic velocity addition	Physics / Relativity / Velocity Addition
I.16.6	$\frac{u+v}{1+\frac{uv}{c^2}}$	$u \in [1.0, 5.0]$	u : velocity of object in frame S'	Relativistic velocity addition	Physics / Relativity / Velocity Addition
I.18.4	$\frac{m_1r_1+m_2r_2}{m_1+m_2}$	$v \in [1.0, 5.0]$ $c \in [1.0, 5.0]$	v : velocity of frame S' relative to S c : speed of light in vacuum	Relativistic velocity addition	Physics / Relativity / Velocity Addition
I.18.4	$\frac{m_1r_1+m_2r_2}{m_1+m_2}$	$m1 \in [1.0, 5.0]$ $m2 \in [1.0, 5.0]$ $r1 \in [1.0, 5.0]$	m_1 : mass of particle 1 m_2 : mass of particle 2 r_1 : position of particle 1	Center of mass position	Physics / Classical Mechanics / Center of Mass

I.18.12	$Fr \sin(\theta)$	$r2 \in [1.0, 5.0]$ $r \in [1.0, 5.0]$	$r2$: position of particle 2 F : force	Torque (moment of force)	Physics / Classical Mechanics / Rotational Dynamics	
I.18.14	$mrv \sin(\theta)$	$F \in [1.0, 5.0]$ $\theta \in [0.0, 5.0]$ $m \in [1.0, 5.0]$	r : distance θ : angle m : mass	Angular momentum	Physics / Classical Mechanics / Angular Momentum	
I.24.6	$\frac{mx^2(\omega^2 + \omega_0^2)}{4}$	$r \in [1.0, 5.0]$ $v \in [1.0, 5.0]$ $\theta \in [1.0, 5.0]$ $m \in [1.0, 3.0]$	r : distance v : velocity θ : angle m : mass	Energy of forced harmonic oscillator	Physics / Classical Mechanics / Harmonic Oscillator	
I.25.13	$\frac{q}{C}$	$\omega \in [1.0, 3.0]$ $\omega_0 \in [1.0, 3.0]$ $x \in [1.0, 3.0]$ $q \in [1.0, 5.0]$	ω : driving frequency ω_0 : natural frequency x : amplitude q : charge	Voltage across capacitor	Physics / Electromagnetism / Capacitance	
I.26.2	$\arcsin(n \sin(\theta_2))$	$C \in [1.0, 5.0]$ $n \in [0.0, 1.0]$ $\theta_2 \in [1.0, 5.0]$	C : capacitance n : refractive index θ_2 : incident angle	Snell's law (refraction angle)	Physics / Optics / Refraction	
I.27.6	$\frac{1}{\frac{n}{d_2} + \frac{1}{d_1}}$	$dI \in [1.0, 5.0]$ $d2 \in [1.0, 5.0]$ $n \in [1.0, 5.0]$	$d1$: object distance $d2$: focal length n : refractive index	Thin lens equation (image distance)	Physics / Optics / Geometric Optics	
I.29.4	$\frac{\omega}{c}$	$\omega \in [1.0, 10.0]$ $c \in [1.0, 10.0]$	ω : angular frequency c : wave speed	Wave number (k)	Physics / Wave Mechanics / Wave Properties	
I.29.16	$\sqrt{x_1^2 - 2x_1x_2 \cos(\theta_1 - \theta_2) + x_2^2}$	$xI \in [1.0, 5.0]$ $x2 \in [1.0, 5.0]$ $\theta_1 \in [1.0, 5.0]$ $\theta_2 \in [1.0, 5.0]$	$x1$: magnitude of vector 1 $x2$: magnitude of vector 2 θ_1 : polar angle of vector 1 θ_2 : polar angle of vector 2	Law of cosines (distance in polar coords)	Mathematics / Geometry / Trigonometry	✓
I.30.3	$\frac{Int_0 \sin^2\left(\frac{n\theta}{2}\right)}{\sin^2\left(\frac{\theta}{2}\right)}$	$Int_0 \in [1.0, 5.0]$ $\theta \in [1.0, 5.0]$ $n \in [1.0, 5.0]$	Int_0 : max intensity θ : angle n : number of slits	Multiple-slit diffraction intensity	Physics / Optics / Diffraction	✓
I.30.5	$\arcsin\left(\frac{\lambda n b d}{d n}\right)$	$\lambda \in [1.0, 2.0]$ $d \in [2.0, 5.0]$ $n \in [1.0, 5.0]$	λ : wavelength d : grating spacing n : order	Diffraction grating angle	Physics / Optics / Diffraction	
I.32.5	$\frac{a^2 q^2}{6\pi c^3 \epsilon}$	$q \in [1.0, 5.0]$ $a \in [1.0, 5.0]$ $\epsilon \in [1.0, 5.0]$ $c \in [1.0, 5.0]$	q : charge a : acceleration ϵ : permittivity c : speed of light	Abraham-Lorentz radiation power	Physics / Electromagnetism / Radiation	
I.32.17	$\frac{4\pi E f^2 c \epsilon \omega^4 r^2}{3(\omega^2 - \omega_0^2)^2}$	$Ef \in [1.0, 2.0]$ $\epsilon \in [1.0, 2.0]$ $c \in [1.0, 2.0]$ $\omega \in [1.0, 2.0]$ $\omega_0 \in [3.0, 5.0]$ $r \in [1.0, 2.0]$	Ef : electric field amplitude ϵ : permittivity of medium c : speed of light ω : incident light frequency ω_0 : resonance frequency r : particle radius	Rayleigh scattering intensity	Physics / Optics / Light Scattering	✓

1.34.8	$\frac{Bqv}{p}$	$B \in [1.0, 5.0]$ $q \in [1.0, 5.0]$ $v \in [1.0, 5.0]$ $p \in [1.0, 5.0]$	B : magnetic field strength q : particle charge v : particle velocity p : particle momentum	Cyclotron frequency	Physics / Electromagnetism / Particle Dynamics
1.34.1	$\frac{\omega_0}{1 - \frac{v}{c}}$	$c \in [3.0, 10.0]$ $v \in [1.0, 2.0]$ $\omega_0 \in [1.0, 5.0]$	ω_0 : rest frequency v : velocity c : speed of light	Doppler shift (approaching)	Physics / Wave Mechanics / Doppler Effect
1.34.14	$\frac{\omega_0 \cdot \left(1 + \frac{v}{c}\right)}{\sqrt{1 - \frac{v^2}{c^2}}}$	$c \in [3.0, 10.0]$ $v \in [1.0, 2.0]$ $\omega_0 \in [1.0, 5.0]$	ω_0 : rest frequency v : velocity c : speed of light	Relativistic Doppler shift	Physics / Relativity / Doppler Effect ✓
1.34.27	$\frac{h\omega}{2\pi}$	$\omega \in [1.0, 5.0]$ $h \in [1.0, 5.0]$	h : Planck's constant ω : angular frequency	Photon energy (quantum mechanics)	Physics / Quantum Mechanics / Photon Energy
1.37.4	$I_1 + I_2 + 2\sqrt{I_1 I_2} \cos(\delta)$	$I1 \in [1.0, 5.0]$ $I2 \in [1.0, 5.0]$ $\delta \in [1.0, 5.0]$	I_1 : intensity of wave 1 I_2 : intensity of wave 2 δ : phase difference between waves	Interference of two waves	Physics / Optics / Wave Interference ✓
1.38.12	$\frac{\epsilon h^2}{\pi m q^2}$	$h \in [1.0, 5.0]$ $m \in [1.0, 5.0]$ $q \in [1.0, 5.0]$ $\epsilon \in [1.0, 5.0]$	h : Planck's constant m : electron mass q : electron charge ϵ : permittivity of vacuum	Bohr radius (atomic physics)	Physics / Quantum Mechanics / Atomic Structure
1.39.1	$\frac{3Vpr}{2}$	$V \in [1.0, 5.0]$ $pr \in [1.0, 5.0]$	V : gas volume pr : gas pressure	Internal energy of monatomic ideal gas	Physics / Thermodynamics / Ideal Gas
1.39.11	$\frac{Vpr}{\gamma - 1}$	$V \in [1.0, 5.0]$ $pr \in [1.0, 5.0]$ $\gamma \in [2.0, 5.0]$	V : gas volume pr : gas pressure γ : heat capacity ratio C_p/C_v	Internal energy of ideal gas	Physics / Thermodynamics / Ideal Gas
1.39.22	$\frac{Tkb n}{V}$	$T \in [1.0, 5.0]$ $kb \in [1.0, 5.0]$ $n \in [1.0, 5.0]$ $V \in [1.0, 5.0]$	T : absolute temperature kb : Boltzmann constant n : number of molecules V : gas volume	Ideal gas law (pressure)	Physics / Thermodynamics / Gas Laws ✓
1.40.1	$n_0 e^{-\frac{gmx}{Tkb}}$	$n_0 \in [1.0, 5.0]$ $m \in [1.0, 5.0]$ $x \in [1.0, 5.0]$ $T \in [1.0, 5.0]$ $g \in [1.0, 5.0]$ $kb \in [1.0, 5.0]$	n_0 : reference density m : molecular mass x : height T : temperature g : gravity kb : Boltzmann constant	Barometric formula (atmospheric pressure)	Physics / Fluid Dynamics / Atmospheric Physics ✓
1.41.16	$\frac{h\omega^3}{2\pi^3 c^2 \left(e^{\frac{h\omega}{2\pi Tkb}} - 1\right)}$	$h \in [1.0, 5.0]$ $\omega \in [1.0, 5.0]$ $T \in [1.0, 5.0]$ $kb \in [1.0, 5.0]$ $c \in [1.0, 5.0]$	h : Planck's constant ω : angular frequency of radiation T : absolute temperature kb : Boltzmann constant c : speed of light	Planck's law (blackbody radiation)	Physics / Quantum Mechanics / Blackbody Radiation ✓

I.43.16	$\frac{Volt \mu_{drift} q}{d}$	$\mu_{drift} \in [1.0, 5.0]$ $q \in [1.0, 5.0]$ $Volt \in [1.0, 5.0]$ $d \in [1.0, 5.0]$	μ_{drift} : drift mobility q : charge $Volt$: voltage d : distance	Drift current	Physics / Solid State Physics / Charge Transport
I.43.31	$T k b m o b$	$T \in [1.0, 5.0]$ $k b \in [1.0, 5.0]$ $m o b \in [1.0, 5.0]$	T : absolute temperature $k b$: Boltzmann constant $m o b$: carrier mobility	Einstein relation (diffusion coefficient)	Physics / Statistical Mechanics / Transport Phenomena
I.43.43	$\frac{k b v}{A(\gamma-1)}$	$k b \in [1.0, 5.0]$ $v \in [1.0, 5.0]$ $A \in [1.0, 5.0]$ $\gamma \in [2.0, 5.0]$	$k b$: Boltzmann constant v : mean molecular velocity A : molecular cross-section γ : heat capacity ratio	Mean free path in gas	Physics / Statistical Mechanics / Kinetic Theory
I.44.4	$T k b n \log \left(\frac{V_2}{V_1} \right)$	$T \in [1.0, 5.0]$ $k b \in [1.0, 5.0]$ $n \in [1.0, 5.0]$ $V_1 \in [1.0, 5.0]$ $V_2 \in [1.0, 5.0]$	T : absolute temperature $k b$: Boltzmann constant n : number of particles V_1 : initial volume V_2 : final volume	Entropy change in isothermal expansion	Physics / Thermodynamics / Entropy ✓
I.47.23	$\sqrt{\frac{\gamma p r}{\rho}}$	$\gamma \in [1.0, 5.0]$ $p r \in [1.0, 5.0]$ $\rho \in [1.0, 5.0]$	γ : heat capacity ratio $p r$: pressure ρ : density	Speed of sound in gas	Physics / Fluid Dynamics / Acoustics
I.48.2	$\frac{c^2 m}{\sqrt{1 - \frac{v^2}{c^2}}}$	$m \in [1.0, 5.0]$ $v \in [1.0, 2.0]$ $c \in [3.0, 10.0]$	m : rest mass v : velocity c : speed of light	Relativistic energy (E=mc ² /y)	Physics / Relativity / Mass-Energy Equivalence
I.50.26	$x_1 \left(\alpha \cos^2(\omega t) + \cos(\omega t) \right)$	$x_1 \in [1.0, 3.0]$ $\omega \in [1.0, 3.0]$ $t \in [1.0, 3.0]$ $\alpha \in [1.0, 3.0]$	x_1 : amplitude ω : frequency t : time α : nonlinearity param	Nonlinear oscillation	Physics / Classical Mechanics / Non-linear Dynamics ✓
II.2.42	$\frac{A \kappa (-T_1 + T_2)}{d}$	$\kappa \in [1.0, 5.0]$ $T_1 \in [1.0, 5.0]$ $T_2 \in [1.0, 5.0]$ $A \in [1.0, 5.0]$ $d \in [1.0, 5.0]$	κ : thermal conductivity T_1 : temperature at cold side T_2 : temperature at hot side A : cross-sectional area d : material thickness	Heat flow (Fourier's law)	Physics / Thermodynamics / Heat Transfer
II.3.24	$\frac{P w r}{4 \pi r^2}$	$P w r \in [1.0, 5.0]$ $r \in [1.0, 5.0]$	$P w r$: total power r : distance	Intensity from point source (inverse square law)	Physics / Wave Mechanics / Intensity
II.4.23	$\frac{q}{4 \pi \epsilon r}$	$q \in [1.0, 5.0]$ $\epsilon \in [1.0, 5.0]$ $r \in [1.0, 5.0]$	q : charge ϵ : permittivity r : distance	Electric potential from point charge	Physics / Electromagnetism / Electric Potential
II.6.11	$\frac{p_d \cos(\theta)}{4 \pi \epsilon r^2}$	$p_d \in [1.0, 3.0]$ $\epsilon \in [1.0, 3.0]$ $r \in [1.0, 3.0]$ $\theta \in [1.0, 3.0]$	p_d : electric dipole moment ϵ : permittivity of medium r : distance from dipole θ : angle from dipole axis	Electric potential from dipole	Physics / Electromagnetism / Dipole Field

II.6.15a	$\frac{3p_d z \sqrt{x^2 + y^2}}{4\pi\epsilon r^5}$	$p_d \in [1.0, 3.0]$ $\epsilon \in [1.0, 3.0]$ $x \in [1.0, 3.0]$ $y \in [1.0, 3.0]$ $z \in [1.0, 3.0]$ $r \in [1.0, 3.0]$	p_d : electric dipole moment ϵ : permittivity of medium x : Cartesian x-coordinate y : Cartesian y-coordinate z : Cartesian z-coordinate r : radial distance from dipole	Electric field component from dipole (cylindrical coords)	Physics / Electromagnetism / Dipole Field ✓
II.6.15b	$\frac{3p_d \sin(\theta) \cos(\theta)}{4\pi\epsilon r^3}$	$p_d \in [1.0, 3.0]$ $\epsilon \in [1.0, 3.0]$ $r \in [1.0, 3.0]$ $\theta \in [1.0, 3.0]$	p_d : electric dipole moment ϵ : permittivity of medium r : radial distance from dipole θ : polar angle from dipole axis	Electric field from dipole (spherical coords)	Physics / Electromagnetism / Dipole Field ✓
II.8.7	$\frac{3q^2}{20\pi d\epsilon}$	$q \in [1.0, 5.0]$ $\epsilon \in [1.0, 5.0]$ $d \in [1.0, 5.0]$	q : charge ϵ : permittivity d : radius	Capacitance of spherical conductor	Physics / Electromagnetism / Capacitance
II.8.31	$\frac{Ef^2\epsilon}{2}$	$Ef \in [1.0, 5.0]$ $\epsilon \in [1.0, 5.0]$	Ef : electric field magnitude ϵ : permittivity of medium	Electric field energy density	Physics / Electromagnetism / Energy Density
II.10.9	$\frac{\sigma_{den}}{\epsilon(\chi+1)}$	$\sigma_{den} \in [1.0, 5.0]$ $\epsilon \in [1.0, 5.0]$ $\chi \in [1.0, 5.0]$	σ_{den} : charge density ϵ : permittivity χ : susceptibility	Electric field in dielectric	Physics / Electromagnetism / Dielectric Materials
II.11.3	$\frac{Ef q}{m(-\omega^2 + \omega_0^2)}$	$Ef \in [1.0, 3.0]$ $q \in [1.0, 3.0]$ $m \in [1.0, 3.0]$ $\omega_0 \in [3.0, 5.0]$ $\omega \in [1.0, 2.0]$	Ef : driving electric field q : particle charge m : particle mass ω_0 : natural frequency ω : driving frequency	Driven harmonic oscillator amplitude	Physics / Classical Mechanics / Forced Oscillation
II.11.17	$n_0 \left(\frac{Ef p_d \cos(\theta)}{T kb} + 1 \right)$	$n_0 \in [1.0, 3.0]$ $Ef \in [1.0, 3.0]$ $p_d \in [1.0, 3.0]$ $T \in [1.0, 3.0]$ $kb \in [1.0, 3.0]$ $\theta \in [1.0, 3.0]$	n_0 : reference number density Ef : applied electric field p_d : molecular dipole moment T : absolute temperature kb : Boltzmann constant θ : angle between field and dipole	Polarization density (Langevin function approximation)	Physics / Electromagnetism / Polarization ✓
II.11.20	$\frac{Ef n_\rho p_d^2}{3T kb}$	$Ef \in [1.0, 5.0]$ $n_\rho \in [1.0, 5.0]$ $p_d \in [1.0, 5.0]$ $T \in [1.0, 5.0]$ $kb \in [1.0, 5.0]$	Ef : applied electric field n_ρ : molecular number density p_d : molecular dipole moment T : absolute temperature kb : Boltzmann constant	Polarization (electric susceptibility)	Physics / Electromagnetism / Polarization ✓
II.11.27	$\frac{Ef \alpha \epsilon n}{-\frac{\alpha n}{3} + 1}$	$Ef \in [1.0, 2.0]$ $n \in [0.0, 1.0]$ $\alpha \in [0.0, 1.0]$ $\epsilon \in [1.0, 2.0]$	Ef : applied electric field n : molecular number density α : molecular polarizability ϵ : permittivity of vacuum	Clausius-Mossotti relation (polarization)	Physics / Electromagnetism / Dielectric Materials ✓
II.11.28	$\frac{\alpha n}{-\frac{\alpha n}{3} + 1} + 1$	$\alpha \in [0.0, 1.0]$ $n \in [0.0, 1.0]$	α : molecular polarizability n : molecular number density	Dielectric constant from polarizability	Physics / Electromagnetism / Dielectric Materials ✓

II.13.17	$\frac{I}{2\pi c^2 \epsilon r}$	$I \in [1.0, 5.0]$ $c \in [1.0, 5.0]$ $\epsilon \in [1.0, 5.0]$ $r \in [1.0, 5.0]$	I : electric current c : speed of light ϵ : permittivity of vacuum r : distance from wire	Magnetic field from current (Biot-Savart)	Physics / Electromagnetism / Magnetic Field
II.13.23	$\frac{\rho_{c0}}{\sqrt{1-\frac{v^2}{c^2}}}$	$\rho_{c0} \in [1.0, 5.0]$ $v \in [1.0, 2.0]$ $c \in [3.0, 10.0]$	ρ_{c0} : rest charge density v : velocity c : speed of light	Relativistic charge density	Physics / Relativity / Charge Density ✓
II.13.34	$\frac{\rho_{c0} v}{\sqrt{1-\frac{v^2}{c^2}}}$	$\rho_{c0} \in [1.0, 5.0]$ $v \in [1.0, 2.0]$ $c \in [3.0, 10.0]$	ρ_{c0} : rest charge density v : velocity c : speed of light	Relativistic current density	Physics / Relativity / Current Density ✓
II.15.4	$-Bmom \cos(\theta)$	$B \in [1.0, 5.0]$ $mom \in [1.0, 5.0]$ $\theta \in [1.0, 5.0]$	B : magnetic field strength mom : magnetic dipole moment θ : angle between field and moment	Magnetic dipole potential energy	Physics / Electromagnetism / Magnetic Dipole
II.15.5	$-Efp_d \cos(\theta)$	$Ef \in [1.0, 5.0]$ $p_d \in [1.0, 5.0]$ $\theta \in [1.0, 5.0]$	Ef : electric field strength p_d : electric dipole moment θ : angle between field and dipole	Electric dipole potential energy	Physics / Electromagnetism / Electric Dipole
II.21.32	$\frac{q}{4\pi\epsilon r \left(1-\frac{v}{c}\right)}$	$q \in [1.0, 5.0]$ $\epsilon \in [1.0, 5.0]$ $r \in [1.0, 5.0]$ $v \in [1.0, 2.0]$ $c \in [3.0, 10.0]$	q : charge ϵ : permittivity r : distance v : velocity c : speed of light	Liénard-Wiechert potential (moving charge)	Physics / Electromagnetism / Retarded Potential
II.24.17	$\sqrt{-\frac{\pi^2}{d^2} + \frac{\omega^2}{c^2}}$	$\omega \in [4.0, 6.0]$ $c \in [1.0, 2.0]$ $d \in [2.0, 4.0]$	ω : frequency c : wave speed d : waveguide dimension	Wave number in waveguide	Engineering / Electrical Engineering / Waveguide Theory ✓
II.27.16	$Ef^2 c \epsilon$	$Ef \in [1.0, 5.0]$ $c \in [1.0, 5.0]$ $\epsilon \in [1.0, 5.0]$	Ef : electric field amplitude c : speed of light ϵ : permittivity of medium	Poynting vector magnitude	Physics / Electromagnetism / Energy Flow
II.27.18	$Ef^2 \epsilon$	$Ef \in [1.0, 5.0]$ $\epsilon \in [1.0, 5.0]$	Ef : electric field amplitude ϵ : permittivity of medium	Electric energy flux	Physics / Electromagnetism / Energy Density
II.34.2a	$\frac{qv}{2\pi r}$	$q \in [1.0, 5.0]$ $v \in [1.0, 5.0]$ $r \in [1.0, 5.0]$	q : charge v : velocity r : radius	Magnetic dipole moment (current loop)	Physics / Electromagnetism / Magnetic Moment
II.34.2	$\frac{qrv}{2}$	$q \in [1.0, 5.0]$ $r \in [1.0, 5.0]$ $v \in [1.0, 5.0]$	q : particle charge r : orbital radius v : orbital velocity	Magnetic moment (orbiting charge)	Physics / Electromagnetism / Magnetic Moment
II.34.11	$\frac{Bg}{2m} q$	$B \in [1.0, 5.0]$ $g \in [1.0, 5.0]$ $q \in [1.0, 5.0]$ $m \in [1.0, 5.0]$	B : magnetic field strength g : Landé g-factor q : particle charge m : particle mass	Larmor precession frequency	Physics / Quantum Mechanics / Spin Precession

✓

II.34.29a	$\frac{h q}{4 \pi m}$	$h \in [1.0, 5.0]$ $q \in [1.0, 5.0]$ $m \in [1.0, 5.0]$	h : Planck's constant q : electron charge m : electron mass	Bohr magneton	Physics / Quantum Mechanics / Magnetic Moment
II.34.29b	$\frac{2 \pi B J z}{h} \frac{mom}{h}$	$B \in [1.0, 5.0]$ $J z \in [1.0, 5.0]$ $z \in [1.0, 5.0]$ $mom \in [1.0, 5.0]$ $h \in [1.0, 5.0]$	B : applied magnetic field $J z$: z-component of angular momentum g : Landé g-factor mom : Bohr magneton h : Planck's constant	Zeeman effect energy splitting	Physics / Quantum Mechanics / Atomic Spectroscopy
II.35.18	$\frac{n_0}{e} \frac{B mom}{T k b} + e^{-\frac{B mom}{T k b}}$	$n_0 \in [1.0, 3.0]$ $B \in [1.0, 3.0]$ $mom \in [1.0, 3.0]$ $T \in [1.0, 3.0]$ $k b \in [1.0, 3.0]$	n_0 : total number density B : applied magnetic field mom : magnetic moment per particle T : absolute temperature $k b$: Boltzmann constant	Paramagnetism (Brillouin function approx)	Physics / Solid State Physics / Magnetism ✓
II.35.21	$mom n_{\rho} \tanh\left(\frac{B mom}{T k b}\right)$	$mom \in [1.0, 5.0]$ $n_{\rho} \in [1.0, 5.0]$ $B \in [1.0, 5.0]$ $T \in [1.0, 5.0]$ $k b \in [1.0, 5.0]$	mom : magnetic moment per particle n_{ρ} : particle number density B : applied magnetic field T : absolute temperature $k b$: Boltzmann constant	Magnetization (Langevin function)	Physics / Solid State Physics / Magnetization ✓
II.36.38	$\frac{H mom}{T k b} + \frac{M mom}{T c^2 \epsilon k b}$	$H \in [1.0, 3.0]$ $M \in [1.0, 3.0]$ $mom \in [1.0, 3.0]$ $T \in [1.0, 3.0]$ $k b \in [1.0, 3.0]$ $\alpha \in [1.0, 3.0]$ $\epsilon \in [1.0, 3.0]$ $c \in [1.0, 3.0]$	H : applied magnetic field M : magnetization mom : magnetic moment per particle T : absolute temperature $k b$: Boltzmann constant α : magneto-electric coupling constant ϵ : permittivity of medium c : speed of light	Magnetization with electric contribution	Physics / Solid State Physics / Magneto-Electric Coupling ✓
II.37.1	$B mom (\chi + 1)$	$B \in [1.0, 5.0]$ $mom \in [1.0, 5.0]$ $\chi \in [1.0, 5.0]$	B : applied magnetic field mom : magnetic moment per particle χ : magnetic susceptibility	Magnetic field in material	Physics / Solid State Physics / Magnetic Materials
II.38.3	$\frac{A Y x}{d}$	$Y \in [1.0, 5.0]$ $A \in [1.0, 5.0]$ $x \in [1.0, 5.0]$ $d \in [1.0, 5.0]$	Y : Young's modulus A : cross-sectional area x : elongation (extension) d : original length	Hooke's law for stress/strain (force)	Engineering / Mechanical Engineering / Elasticity
II.38.14	$\frac{Y}{2 \sigma + 2}$	$Y \in [1.0, 5.0]$ $\sigma \in [1.0, 5.0]$	Y : Young's modulus σ : Poisson's ratio	Shear modulus from Young's modulus	Engineering / Mechanical Engineering / Material Properties
III.4.32	$\frac{1}{e^{\frac{h \omega}{2 \pi T k b}} - 1}$	$h \in [1.0, 5.0]$ $\omega \in [1.0, 5.0]$ $T \in [1.0, 5.0]$ $k b \in [1.0, 5.0]$	h : Planck's constant ω : angular frequency T : absolute temperature $k b$: Boltzmann constant	Bose-Einstein distribution	Physics / Statistical Mechanics / Quantum Statistics
III.4.33	$\frac{h \omega}{2 \pi \left(e^{\frac{h \omega}{2 \pi T k b}} - 1 \right)}$	$h \in [1.0, 5.0]$ $\omega \in [1.0, 5.0]$ $T \in [1.0, 5.0]$	h : Planck's constant ω : angular frequency T : absolute temperature	Average energy (Bose-Einstein statistics)	Physics / Statistical Mechanics / Quantum Statistics ✓

III.7.38	$\frac{4\pi B mom}{h}$	$kb \in [1.0, 5.0]$ $B \in [1.0, 5.0]$ $mom \in [1.0, 5.0]$ $h \in [1.0, 5.0]$	kb : Boltzmann constant B : magnetic field mom : magnetic dipole moment h : Planck's constant	Gyromagnetic ratio times field	Physics / Quantum Mechanics / Spin Dynamics
III.8.54	$\sin^2\left(\frac{2\pi E_n t}{h}\right)$	$E_n \in [1.0, 2.0]$ $t \in [1.0, 2.0]$ $h \in [1.0, 4.0]$	E_n : energy t : time h : Planck's constant	Quantum transition probability (time-dependent)	Physics / Quantum Mechanics / Time Evolution
III.9.52	$\frac{8\pi Ef p_d \sin^2\left(\frac{t(\omega-\omega_0)}{2}\right)}{ht(\omega-\omega_0)^2}$	$Ef \in [1.0, 3.0]$ $p_d \in [1.0, 3.0]$ $\omega \in [1.0, 5.0]$ $\omega_0 \in [1.0, 5.0]$ $t \in [1.0, 3.0]$ $h \in [1.0, 3.0]$	Ef : electric field amplitude p_d : electric dipole moment ω : driving frequency ω_0 : resonance frequency t : time h : Planck's constant	Rabi oscillation transition rate	Physics / Quantum Mechanics / Quantum Optics ✓
III.10.19	$mom\sqrt{Bx^2 + By^2 + Bz^2}$	$mom \in [1.0, 5.0]$ $Bx \in [1.0, 5.0]$ $By \in [1.0, 5.0]$ $Bz \in [1.0, 5.0]$	mom : magnetic dipole moment Bx : x-component of magnetic field By : y-component of magnetic field Bz : z-component of magnetic field	Energy in 3D magnetic field	Physics / Quantum Mechanics / Zeeman Energy ✓
III.12.43	$\frac{\hbar n}{2\pi}$	$h \in [1.0, 5.0]$ $n \in [1.0, 5.0]$	h : Planck's constant n : principal quantum number	Quantized angular momentum	Physics / Quantum Mechanics / Angular Momentum
III.13.18	$\frac{4\pi E_n d^2 k}{h}$	$E_n \in [1.0, 5.0]$ $d \in [1.0, 5.0]$ $k \in [1.0, 5.0]$ $h \in [1.0, 5.0]$	E_n : energy d : matrix element k : density of states h : Planck's constant	Fermi's golden rule (transition rate)	Physics / Quantum Mechanics / Perturbation Theory
III.14.14	$I_0 \left(e^{\frac{Volt q}{T kb}} - 1 \right)$	$I_0 \in [1.0, 5.0]$ $Volt \in [1.0, 2.0]$ $q \in [1.0, 2.0]$ $T \in [1.0, 2.0]$ $kb \in [1.0, 2.0]$ $U \in [1.0, 5.0]$	I_0 : saturation current $Volt$: applied voltage q : elementary charge T : absolute temperature kb : Boltzmann constant	Shockley diode equation	Engineering / Electrical Engineering / Semiconductor Devices
III.15.12	$2U(1 - \cos(dk))$	$U \in [1.0, 5.0]$ $k \in [1.0, 5.0]$ $d \in [1.0, 5.0]$	U : coupling energy k : wave vector d : lattice constant	Tight-binding dispersion relation	Physics / Solid State Physics / Band Structure
III.15.14	$\frac{\hbar^2}{8\pi^2 E_n d^2}$	$h \in [1.0, 5.0]$ $E_n \in [1.0, 5.0]$ $d \in [1.0, 5.0]$	h : Planck's constant E_n : energy d : well width	Effective mass in quantum well	Physics / Quantum Mechanics / Quantum Confinement
III.15.27	$\frac{2\pi\alpha}{dn}$	$\alpha \in [1.0, 5.0]$ $n \in [1.0, 5.0]$ $d \in [1.0, 5.0]$	α : lattice parameter n : Miller index d : spacing	Reciprocal lattice vector	Physics / Solid State Physics / Crystallography
III.17.37	$\beta(\alpha \cos(\theta) + 1)$	$\beta \in [1.0, 5.0]$ $\alpha \in [1.0, 5.0]$ $\theta \in [1.0, 5.0]$	β : amplitude parameter α : modulation depth θ : angle	Angular modulation function	Mathematics / Applied Mathematics / Signal Processing

III.19.51	$-\frac{mq^4}{8\epsilon^2 h^2 n^2}$	$m \in [1.0, 5.0]$ $q \in [1.0, 5.0]$ $h \in [1.0, 5.0]$ $n \in [1.0, 5.0]$ $\epsilon \in [1.0, 5.0]$	m : electron mass q : electron charge h : Planck's constant n : principal quantum number ϵ : permittivity	Hydrogen atom energy levels (Bohr model)	Physics / Quantum Mechanics / Atomic Physics	
III.21.20	$-\frac{A_{vec} q \rho_{c0}}{m}$	$A_{vec} \in [1.0, 5.0]$ $q \in [1.0, 5.0]$ $\rho_{c0} \in [1.0, 5.0]$ $m \in [1.0, 5.0]$	A_{vec} : magnetic vector potential q : electric charge ρ_{c0} : charge density m : particle mass	Current density in quantum mechanics	Physics / Quantum Mechanics / Electromagnetic Coupling	✓
Bonus1.0	$\frac{Z_1^2 Z_2^2 \alpha^2 c^2 h^2}{16 E_n^2 \sin^4\left(\frac{\theta}{2}\right)}$	$Z_1 \in [1.0, 2.0]$ $Z_2 \in [1.0, 2.0]$ $\alpha \in [1.0, 2.0]$ $h \in [1.0, 2.0]$ $c \in [1.0, 2.0]$ $E_n \in [1.0, 3.0]$ $\theta \in [1.0, 3.0]$	Z_1 : atomic number of projectile Z_2 : atomic number of target α : fine structure constant h : reduced Planck constant c : speed of light E_n : kinetic energy of projectile θ : scattering angle	Mott scattering cross-section (relativistic Rutherford)	Physics / Nuclear Physics / Scattering Theory	✓
Bonus2.0	$\frac{k_G m \left(\sqrt{\frac{2E_n L^2}{k_G^2 m}} + 1 \cos(\theta_1 - \theta_2) + 1 \right)}{L^2}$	$m \in [1.0, 3.0]$ $k_G \in [1.0, 3.0]$ $L \in [1.0, 3.0]$ $E_n \in [1.0, 3.0]$ $\theta_1 \in [0.0, 6.0]$ $\theta_2 \in [0.0, 6.0]$	m : reduced mass of system k_G : gravitational coupling L : total angular momentum E_n : total mechanical energy θ_1 : current true anomaly θ_2 : periapsis argument	Effective potential in two-body problem	Physics / Classical Mechanics / Central Force Motion	✓
Bonus3.0	$\frac{d(1-\alpha^2)}{\alpha \cos(\theta_1 - \theta_2) + 1}$	$d \in [1.0, 3.0]$ $\alpha \in [2.0, 4.0]$ $\theta_1 \in [4.0, 5.0]$ $\theta_2 \in [4.0, 5.0]$	d : semi-latus rectum of orbit α : orbital eccentricity θ_1 : current true anomaly θ_2 : argument of periapsis	Orbit equation (conic section with eccentricity)	Physics / Classical Mechanics / Orbital Mechanics	✓
Bonus4.0	$\sqrt{2} \sqrt{\frac{E_n - \frac{L^2}{2mr^2} - U}{m}}$	$m \in [1.0, 3.0]$ $E_n \in [8.0, 12.0]$ $U \in [1.0, 3.0]$ $L \in [1.0, 3.0]$ $r \in [1.0, 3.0]$	m : mass E_n : total energy U : potential energy L : angular momentum r : radial distance	Radial velocity (classical orbital mechanics)	Physics / Classical Mechanics / Orbital Dynamics	✓
Bonus5.0	$\frac{2\pi d^{\frac{3}{2}}}{\sqrt{G(m_1 + m_2)}}$	$d \in [1.0, 3.0]$ $G \in [1.0, 3.0]$ $m_1 \in [1.0, 3.0]$ $m_2 \in [1.0, 3.0]$	d : semi-major axis of orbit G : gravitational constant m_1 : mass of primary body m_2 : mass of orbiting body	Kepler's third law (orbital period)	Astronomy / Celestial Mechanics / Keplerian Orbits	✓
Bonus6.0	$\sqrt{\frac{2E_n L^2 \epsilon^2}{Z_1^2 Z_2^2 m q^4}} + 1$	$\epsilon \in [1.0, 3.0]$ $L \in [1.0, 3.0]$ $E_n \in [1.0, 3.0]$ $Z_1 \in [1.0, 3.0]$	ϵ : permittivity of vacuum L : orbital angular momentum E_n : total mechanical energy Z_1 : charge number of particle 1	Eccentricity of Coulomb/Kepler orbit	Physics / Classical Mechanics / Central Force Orbits	✓

		$Z_2 \in [1.0, 3.0]$ $m \in [1.0, 3.0]$ $q \in [1.0, 3.0]$	Z_2 : charge number of particle 2 m : reduced mass of system q : elementary charge		
Bonus7.0	$\sqrt{\frac{8\pi G\rho}{3} - \frac{\alpha c^2}{d^2}}$	$G \in [1.0, 3.0]$ $\rho \in [1.0, 3.0]$ $\alpha \in [1.0, 2.0]$ $c \in [1.0, 2.0]$ $d \in [1.0, 3.0]$	G : grav constant ρ : mass density α : curvature constant c : speed of light d : scale factor	Friedmann equation (Hubble parameter, $H = \dot{a}/a$)	Astronomy / Cosmology / Expansion Dynamics ✓
Bonus8.0	$\frac{E_n}{\frac{E_n(1-\cos(\theta))}{c^2 m} + 1}$	$E_n \in [1.0, 3.0]$ $m \in [1.0, 3.0]$ $c \in [1.0, 3.0]$ $\theta \in [1.0, 3.0]$	E_n : photon energy m : electron mass c : speed of light θ : scattering angle	Compton scattering (scattered photon energy)	Physics / Quantum Mechanics / Photon Scattering
Bonus9.0	$-\frac{32G^4 m_1^2 m_2^2 (m_1 + m_2)}{5c^5 r^5}$	$G \in [1.0, 2.0]$ $c \in [1.0, 2.0]$ $m1 \in [1.0, 5.0]$ $m2 \in [1.0, 5.0]$ $r \in [1.0, 2.0]$	G : gravitational constant c : speed of light m_1 : mass of first body m_2 : mass of second body r : orbital separation	Gravitational wave power loss (binary system)	Physics / General Relativity / Gravitational Radiation ✓
Bonus10.0	$\arccos\left(\frac{\cos(\theta_2) - \frac{v}{c}}{1 - \frac{v \cos(\theta_2)}{c}}\right)$	$c \in [4.0, 6.0]$ $v \in [1.0, 3.0]$ $\theta_2 \in [1.0, 3.0]$	c : speed of light v : velocity θ_2 : angle in rest frame	Relativistic aberration angle	Physics / Relativity / Light Aberration ✓
Bonus11.0	$\frac{4I_0 \sin^2\left(\frac{\alpha}{2}\right) \sin^2\left(\frac{\delta n}{2}\right)}{\alpha^2 \sin^2\left(\frac{\delta}{2}\right)}$	$I_0 \in [1.0, 3.0]$ $\alpha \in [1.0, 3.0]$ $\delta \in [1.0, 3.0]$ $n \in [1.0, 2.0]$	I_0 : max intensity α : single-slit phase δ : inter-slit phase n : number of slits	Fraunhofer diffraction (N-slit interference pattern)	Physics / Optics / Diffraction Patterns ✓
Bonus12.0	$q \left(\frac{4\pi V_0 l t d \epsilon - \frac{dq y^3}{(-d^2 + y^2)^2}}{4\pi \epsilon y^2} \right)$	$q \in [1.0, 5.0]$ $Volt \in [1.0, 5.0]$ $d \in [4.0, 6.0]$ $\epsilon \in [1.0, 5.0]$ $y \in [1.0, 3.0]$	q : point charge $Volt$: potential reference d : sphere radius ϵ : permittivity of vacuum y : distance from sphere center	Electric field from grounded conducting sphere (image charge method)	Physics / Electromagnetism / Electrostatic Boundary Problems ✓
Bonus13.0	$\frac{q}{4\pi\epsilon\sqrt{d^2 - 2dr\cos(\alpha) + r^2}}$	$q \in [1.0, 5.0]$ $\epsilon \in [1.0, 5.0]$ $d \in [4.0, 6.0]$ $r \in [1.0, 3.0]$ $\alpha \in [0.0, 6.0]$	q : point charge magnitude ϵ : permittivity of medium d : distance from origin to charge r : distance from origin to field point α : angle between d and r	Coulomb potential (law of cosines form for arbitrary geometry)	Physics / Electromagnetism / Electric Potential ✓
Bonus14.0	$Ef \left(\frac{d^3(\alpha-1)}{r^2(\alpha+2)} - r \right) \cos(\theta)$	$Ef \in [1.0, 5.0]$ $\alpha \in [1.0, 5.0]$ $d \in [1.0, 5.0]$ $r \in [1.0, 5.0]$ $\theta \in [0.0, 6.0]$	Ef : external electric field α : relative dielectric constant d : sphere radius r : distance from sphere center θ : polar angle from field direction	Induced dipole potential (dielectric sphere in uniform field)	Physics / Electromagnetism / Dielectric Response ✓

Bonus15.0	$\frac{\omega \sqrt{1 - \frac{v^2}{c^2}}}{1 + \frac{v \cos(\theta)}{c}}$	$\omega \in [1.0, 5.0]$ $v \in [1.0, 3.0]$ $c \in [5.0, 20.0]$ $\theta \in [0.0, 6.0]$	ω : source frequency (rest frame) v : relative velocity c : speed of light θ : angle between velocity and observation	Relativistic Doppler shift (general an- Physics / Relativity / Doppler Effect ✓ gle)
Bonus16.0	$Volt q + \sqrt{c^4 m^2 + c^2 (-A_{vec} q + p)^2}$	$Volt \in [1.0, 5.0]$ $q \in [1.0, 5.0]$ $c \in [1.0, 5.0]$ $m \in [1.0, 5.0]$ $A_{vec} \in [1.0, 5.0]$ $p \in [1.0, 5.0]$	$Volt$: scalar potential q : particle charge c : speed of light m : rest mass A_{vec} : magnetic vector potential p : canonical momentum	Relativistic Hamiltonian (charged Physics / Relativity / Relativistic Dy- ✓ particle in EM field, minimal cou- namics pling)
Bonus17.0	$\frac{m^2 \omega^2 x^2 \left(\frac{\alpha x}{y} + 1 \right) + p^2}{2m}$	$m \in [1.0, 5.0]$ $\omega \in [1.0, 5.0]$ $p \in [1.0, 5.0]$ $x \in [1.0, 5.0]$ $\alpha \in [1.0, 5.0]$ $y \in [1.0, 5.0]$	m : particle mass ω : natural angular frequency p : canonical momentum x : displacement from equilibrium α : anharmonicity parameter y : characteristic length scale	Hamiltonian (anharmonic oscillator Physics / Classical Mechanics / ✓ with cubic nonlinearity) Hamiltonian Mechanics
Bonus18.0	$\frac{3 \left(H_G^2 + \frac{c^2 k_f}{r^2} \right)}{8\pi G}$	$H_G \in [1.0, 5.0]$ $c \in [1.0, 5.0]$ $k_f \in [1.0, 5.0]$ $r \in [1.0, 5.0]$ $G \in [1.0, 5.0]$	H_G : Hubble parameter c : speed of light k_f : spatial curvature constant r : scale factor $a(t)$ G : gravitational constant	Friedmann equation (total energy den- Astronomy / Cosmology / Energy ✓ sity from curvature) Density
Bonus19.0	$-\frac{H_G^2 c^2 \cdot (1 - 2\alpha) + \frac{c^4 k_f}{r^2}}{8\pi G}$	$H_G \in [1.0, 5.0]$ $c \in [1.0, 5.0]$ $\alpha \in [1.0, 5.0]$ $k_f \in [1.0, 5.0]$ $r \in [1.0, 5.0]$ $G \in [1.0, 5.0]$	H_G : Hubble parameter c : speed of light α : equation of state parameter w k_f : spatial curvature constant r : scale factor $a(t)$ G : gravitational constant	Friedmann equation (effective pres- Astronomy / Cosmology / Equation of ✓ sure, equation of state parameter $w =$ State α)
Bonus20.0	$\frac{\alpha^2 h^2 \omega_0^2 \left(\frac{\omega}{\omega_0} - \sin^2(\beta) + \frac{\omega_0}{\omega} \right)}{4\pi c^2 m^2 \omega^2}$	$\alpha \in [1.0, 5.0]$ $h \in [1.0, 5.0]$ $\omega_0 \in [1.0, 5.0]$ $\omega \in [1.0, 5.0]$ $m \in [1.0, 5.0]$ $c \in [1.0, 5.0]$ $\beta \in [0.0, 6.0]$	α : fine structure constant h : Planck's constant ω_0 : incident photon frequency ω : scattered photon frequency m : electron rest mass c : speed of light β : scattering angle	Klein-Nishina formula (quantum Physics / Quantum Mechanics / Scat- ✓ Compton differential cross-section) tering Cross-Section

Table 1: Equations.